

A Comparative Study of Machine Learning Regression Algorithms for Predicting AI and Data-Industry Salaries

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Abstract

Finding the amount of competitive compensation is a tough problem for employers, job seekers, and career guidance sites in the fast-growing AI and data professions market. This paper shows a comparison of five supervised machine learning regression algorithms: Linear Regression, Ridge Regression, Decision Tree Regressor, Support Vector Regression (SVR), and Multi-Layer Perceptron (MLP) Regressor, which are used to predict salary in US dollars for AI, machine learning, and data professionals. We used a weekly updated salary survey dataset from ai-jobs.net that contains **71,913 records** from **2020 to 2025**. We cleaned the data by removing duplicates, grouping categories, converting categories to one-hot variables, and scaling features. All models were tested with an 80/20 train-test split and 5-fold cross-validation. We reported MAE, MSE, RMSE, and R2. A special comparison looked at how feature scaling changes results. Standardising data made the MLP perform better (test R2 went from 0.007 to 0.215). The Decision Tree, which does not care about scale, stayed the same. The linear models, which already handle scale well, also stayed the same. SVR saw a small improvement.

After we tuned the hyperparameter α to 1.0, Ridge Regression had the best overall test results, with R2 of 0.220 and an MAE of \$50,993. Removing features one by one showed that job role family and company location are the two useful predictors. Experience level alone only explains an R2 of 0.105 of salary differences. These results mean that for this dataset, regularised linear models give a balance of accuracy and simplicity. Also, feature scaling is necessary, not optional, when using kernel-based regressors on this type of data.

Keywords: Salary prediction, machine learning, regression, feature scaling, Ridge regression, support vector regression, neural networks, comparative study

I. Introduction

The demand for Artificial Intelligence (AI), Machine Learning (ML), and data professionals has gone up quickly since 2020, with often unclear differences in how much people get paid based on their jobs, how much experience they have, how big the companies are, and where they live. Both the people who want to hire others and the people who are trying to get a job can use salary estimates that are based on data. Machine learning regression is a way to do this. When given some details about a job, a model that has been trained can say what the salary might be and also explain which details affect that salary the most.

This project looks at estimating salaries for AI and data jobs as a problem where we try to predict a number. It compares five regression methods that are used a lot. It doesn't just look at which method is the most accurate. It also tries to understand why some methods work better than others. It does this by looking at one choice in how we prepare the data. Whether to scale the features. And by seeing how much each group of features helps with the accuracy. The project follows the steps of a machine learning project. It starts by defining the problem. Then it looks at what others have written about the topic. It chooses a dataset. Explains it. It works on preparing the data, looks at the data to find patterns, tries models, checks one choice about preparing the data in a controlled way, measures how well each model works, and then talks about what the results mean and what the limits are.

A. Research Question

This study addresses the following research question: *Which machine learning regression algorithm most accurately predicts the annual USD salary of an AI/data professional from job- and employment-related attributes, and does feature scaling improve the accuracy of scale-sensitive algorithms (SVR, MLP) relative to a scale-invariant algorithm (Decision Tree)?*

II. Related Work

Several recent studies have looked at using machine learning to predict salaries in tech and data-related jobs. One study used labor-market data from 2020 to 2024. Compared three models: Decision Tree, Random Forest, and Gradient Boosting. It found that the Decision Tree Regressor had the mean squared error among the models tested [1]. Another study compared seven supervised models for predicting employee salaries, looking at both demographic and job-related features [2]. Jiang focused on salary trends in the data-science field using regression-based models [3]. Feng and colleagues compared learning models with traditional machine learning methods and found that conventional models still performed well on tabular, moderately sized datasets [4]. Aminu and team used Principal Component Analysis with a neural network and showed that reducing the number of features helped neural networks when dealing with high-cardinality categorical data [5]. Eichinger and Mayer looked specifically at Random Forest regression. Pointed out how sensitive tree ensembles can be to how features are encoded [6].

On a note another large-scale study looked at how different models respond to feature scaling. It found that models relying on distance or gradients Like Logistic Regression, SVM, k-NN, MLP, and TabNet. Are very sensitive to the choice of scaler. In contrast, tree-based models such as Random Forest and Gradient Boosting are not affected much by scaling [7]. This insight shaped the design of the study presented in this paper as described in Section V. Also, Chicco and team argued that R^2 is a more meaningful metric than MAE, MAPE, MSE or RMSE

alone when evaluating regression models [8]. That’s why this paper reports R^2 along with the three error-based metrics needed for an evaluation.

A common issue in this area of research is that most studies either keep one preprocessing method fixed and change the algorithm or fix the algorithm and vary only the preprocessing. They don’t test both factors together in a way. For example Quan and Raheem studied human-resource analytics for data-science jobs using skill-set features. Did not test how scaling affects different model families [10]. Similarly several studies on graduate salaries and employee satisfaction used Decision Tree, k-means and association-rule methods. Found strong performance from tree-based models but they didn’t explain why linear or neural models didn’t perform as well [11, 12].

Compared to this work this study makes three contributions. First it measures the effect of scaling in a controlled way testing both scaled versions, for every algorithm instead of using just one default setup. Second it does a feature-subset ablation to see how much each categorical group. Like role family, company location and company size. Adds to prediction accuracy. Third it fully discloses the feature engineering and leakage avoidance steps (Section III-B) which are often left out in studies even though they strongly affect accuracy.

III. Dataset

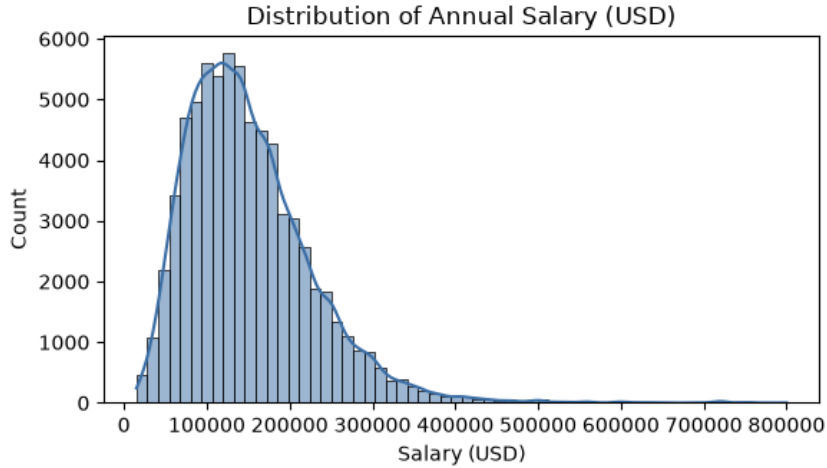
The dataset used in this study, `ai_jobs_salaries_clean.csv`, is derived from the publicly available, weekly-updated AI/ML/Big-Data salary survey published under a CC0 (public-domain) license by `ai-jobs.net` and mirrored on GitHub [9]. Respondents anonymously self-report role, experience, employment type, remote-work arrangement, company size/location and gross salary.

A. Instances and Features

The raw file contains 71,913 instances and 16 candidate features plus the target variable. Table 1 summarises the schema.

Table 1. Dataset schema summary

Property	
Instances	71,913
Raw features	16 (+ 1 target)
Target	salary_in_usd (continuous)
Missing values	40,540, confined to isco_group_hint (56% missing)
Duplicate rows	0
Numeric features	work_year, remote_ratio, salary, salary_in_usd
Categorical features	experience_level, employment_type, job_title (422 levels), company_size, company_location (97 levels), role_family (11 levels)

**Figure 1. Distribution of annual salary (USD).**

The target, salary_in_usd, is right-skewed (skewness = 1.48) with mean \$151,161, median \$138,750 and standard deviation \$77,330, ranging from \$15,000 to \$800,000.

B. Feature Selection

Out of the 16 fields, several were removed before modeling to prevent target leakage and redundancy. The salary and salary_currency fields are the values used to calculate salary_in_usd, so they were excluded. The salary_outlier_flag was dropped because it is directly based on the target

variable and would leak information. The experience_level_label and employment_type_label exactly repeat the data in experience_level and employment_type so they were not kept. The work_mode field duplicates the details in remote_ratio, so it was left out. The job_title field has 422 values and is noisier compared to the pre-grouped role_family, so it was not included. The isco_group_hint was removed due to missing data in 56% of cases, with no way to fill in those gaps. The final feature set for the model includes: experience_level, employment_type, remote_ratio, company_size, role_family, and company_location. For company_location, rare locations not among the eight common countries are grouped into an "Other" category to keep the one-hot encoding manageable.

IV. Methodology

A. Preprocessing

No duplicates were found. Rows with a missing target were removed (none were found). Because Support Vector Regression scales poorly with sample size, a stratified random subsample of 10,000 records (stratified by role_family to preserve the class balance) was drawn from the cleaned 71,913-row dataset for all modelling experiments; this is a standard practice to keep kernel-based methods computationally tractable and is disclosed here as a methodological limitation. All five categorical features were one-hot encoded (with the first level dropped to avoid collinearity), producing 28 engineered features. Numeric features (remote_ratio) were optionally standardised (zero mean, unit variance) depending on the experimental condition (Section V). Data were split 80/20 into training and test sets (seed fixed at 42 throughout), and 5-fold cross-validation was additionally performed on the training set for every model.

B. Exploratory Data Analysis

Salary increases in a way as someone gains more experience, from entry-level up to executive positions. This is shown in Figure 2 where the correlation between salary and experience level's 0.32. Salary also changes a lot depending on the type of job. This is shown in Figure Among the eight common job types, Research Scientist and ML Engineer roles have

the highest median salaries. Data Analyst roles have the lowest median salaries.

The amount of work does not have a strong relationship with salary. The correlation is -0.02 which is zero. This suggests that remote work options are offered in a way across different pay levels, in this sample.

Figure 2. Salary distribution by experience level (EN=Entry, MI=Mid, SE=Senior, EX=Executive).

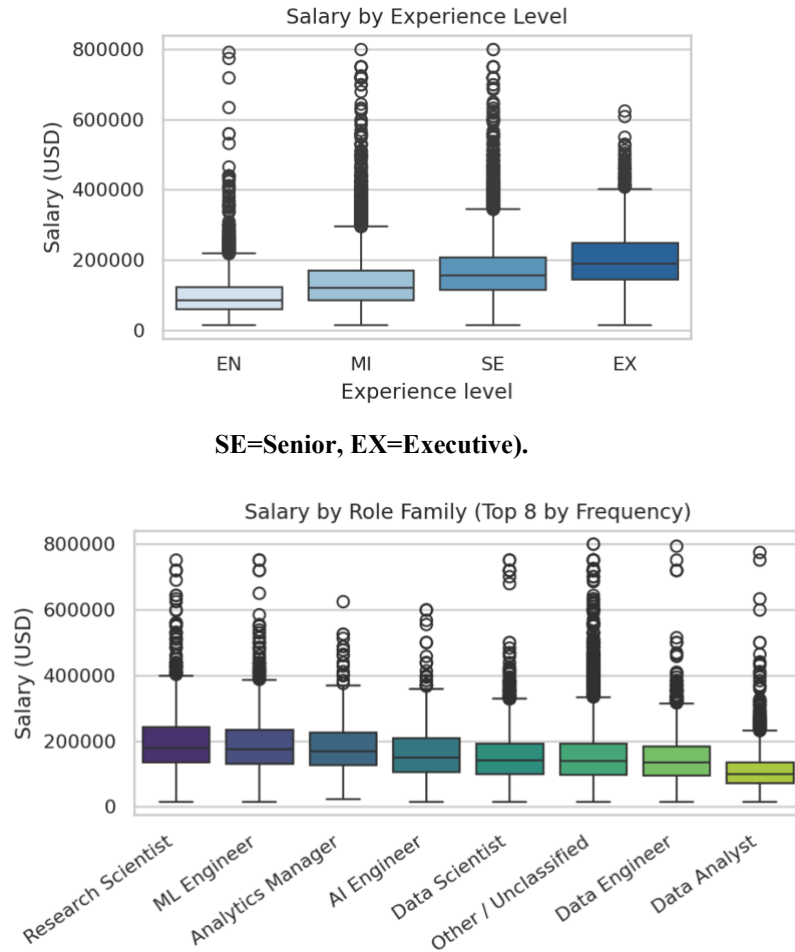
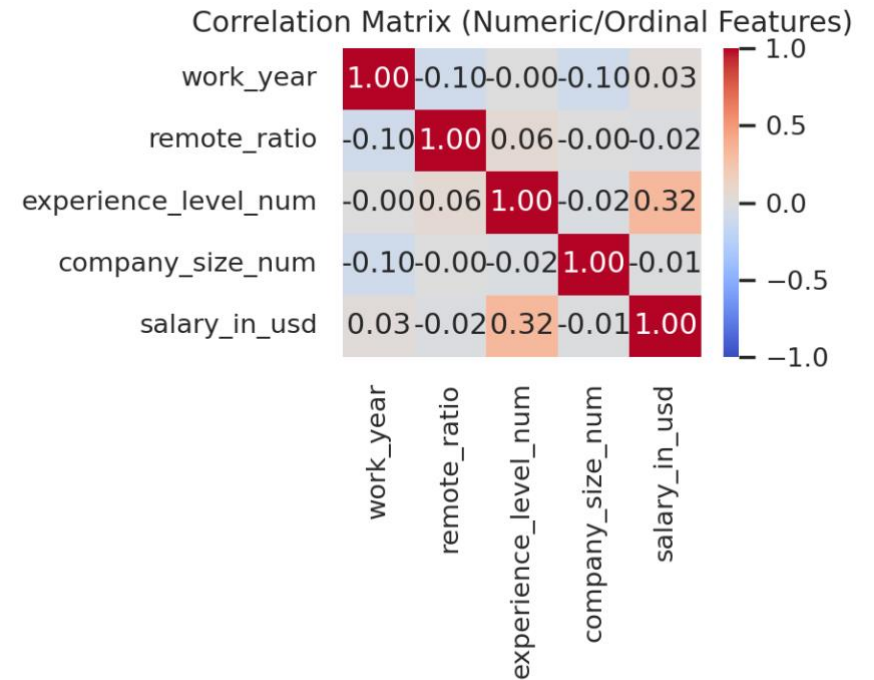


Figure 3. Salary distribution by role family (top 8 by frequency).

Fig. 4 summarises the correlation matrix for the numeric/ordinal features. Experience level is the strongest numeric-style correlate of salary ($r = 0.32$), while company size and remote ratio show weak correlation with salary in isolation. This is consistent with the feature-subset ablation in Section VI-D, which shows that most of the remaining explanatory power comes from the categorical role and location features rather than from these numeric/ordinal fields.

Figure 4. Correlation matrix of numeric/ordinal features and target.



C. Algorithms

Four regression algorithms are required by the project brief; this study implements five to strengthen the comparison: (1) Linear Regression, an ordinary least-squares baseline; (2) Ridge Regression, an L2-regularised linear model (α tuned by grid search); (3) a Decision Tree Regressor (max

depth tuned by grid search); (4) Support Vector Regression with an RBF kernel; and (5) an MLP Regressor (two hidden layers, 64 and 32 units, early stopping). All models were implemented using scikit-learn 1.8 with a fixed random seed of 42.

V. Experimental Setup

A. Comparative Study: Feature Scaling

The required comparative study investigates feature scaling (assignment option A). Every algorithm was trained twice on identical train/test splits: once with the two numeric features (work_year, remote_ratio) left in their raw units, and once with those same two features standardised via StandardScaler. The one-hot encoded categorical columns (already binary 0/1) were left unchanged in both conditions, isolating the scaling effect to the numeric features only.

B. Hyperparameter Tuning

For Ridge Regression, Decision Tree and SVR, a 5-fold GridSearchCV (scoring = negative MAE) was used to select: Ridge $\alpha \in \{0.01, 0.1, 1, 10, 100\}$; Decision Tree $\text{max_depth} \in \{4, 6, 8, 10, 14, \text{None}\}$ and $\text{min_samples_leaf} \in \{1, 5, 10\}$; and SVR $C \in \{1, 10, 50, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.1\}$, $\epsilon \in \{0.1, 1.0\}$.

C. Feature-Subset Ablation

To measure the marginal value of individual feature groups, the tuned Ridge model was retrained five times: with all features; with role_family removed; with company_location removed; with company_size removed; and using experience_level alone.

D. Evaluation Metrics

As this is a regression task, models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2), consistent with the project's regression-evaluation requirement.

VI. Results

A. Baseline Model Comparison

Table 2 reports held-out test-set performance for all five algorithms with default hyperparameters (scale-sensitive models trained on standardised numeric features). Ridge Regression achieved the best test R^2 (0.220) and the lowest MAE (\$50,993), narrowly ahead of Linear Regression and the Decision Tree. SVR performed worst, with a negative R^2 (-0.026), indicating it fit the test data worse than a horizontal mean-prediction line under its default hyperparameters. Five-fold cross-validation (Table 3) confirms the same ranking, with Ridge and Linear Regression producing the highest and most stable mean CV R^2 (0.215, $\text{std} \approx 0.020$).

Table 2. Test-set performance of all algorithms

Model	MAE (\$)	RMSE (\$)	R^2
Linear Regression	51,011	72,032	0.219
Ridge Regression	50,993	72,003	0.220
Decision Tree	51,197	72,161	0.216
SVR (RBF)	58,644	82,543	-0.026
MLP Regressor	51,309	72,232	0.215

Table 3. 5-fold cross-validation results (training set)

Model	Mean R^2	Std R^2	Mean MAE (\$)
Linear Regression	0.214	0.021	50,114
Ridge Regression	0.215	0.020	50,105
Decision Tree	0.196	0.019	50,552
SVR (RBF)	-0.021	0.006	57,171
MLP Regressor	0.208	0.018	50,443

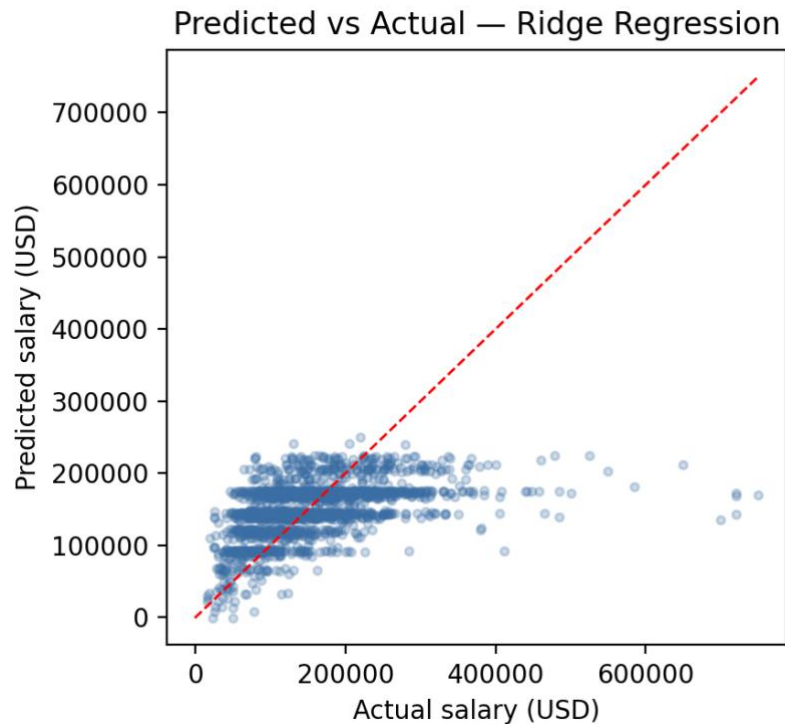


Figure 5. Predicted vs. actual salary for the best-performing model (Ridge Regression).

B. Comparative Study: Effect of Feature Scaling

Table 4 and Fig. 6 present the unscaled-vs-scaled comparison for every algorithm. The effect is highly model-dependent: Linear Regression, Ridge Regression and the Decision Tree are essentially unaffected by scaling the two numeric features (R^2 unchanged to three decimal places), consistent with the scale-invariance of ordinary least squares/L2-regularised linear models and axis-aligned decision trees. SVR improves only marginally (R^2 from -0.035 to -0.026). The MLP Regressor, in contrast, is dramatically affected: test R^2 rises from 0.007 (unscaled) to 0.215 (scaled), and MAE falls from \$59,818 to \$51,309 — a result

consistent with the known sensitivity of gradient-based neural network training to the relative magnitude of input features [7].

Table 4. Effect of feature scaling on test R^2

Model	Unscaled R^2	Scaled R^2
Linear Regression	0.219	0.219
Ridge Regression	0.220	0.220
Decision Tree	0.216	0.216
SVR (RBF)	-0.035	-0.026
MLP Regressor	0.007	0.215

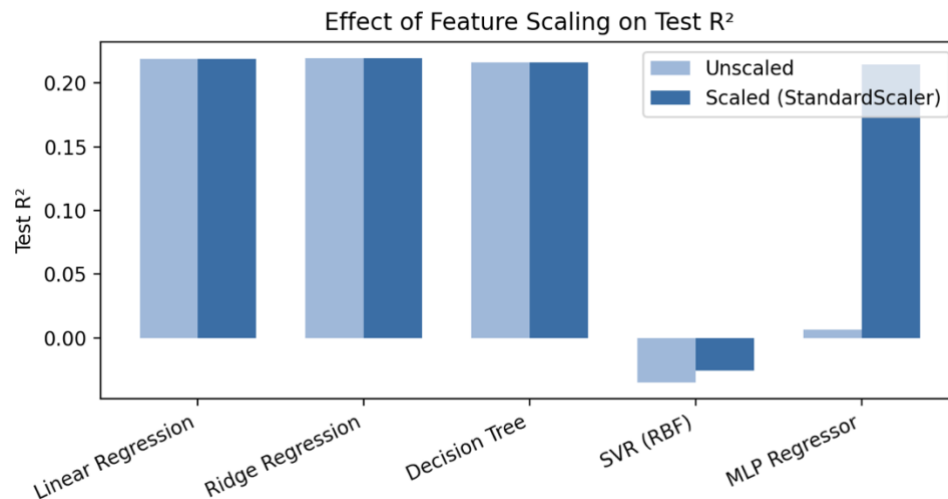


Figure 6. Test R^2 with and without feature scaling, by algorithm.

C. Hyperparameter Tuning

Grid search selected Ridge $\alpha = 1.0$ (the default), Decision Tree $\text{max_depth} = 8$, $\text{min_samples_leaf} = 5$, and SVR $C = 100$, $\gamma = \text{scale}$, $\epsilon = 1.0$. Tuning improved SVR from $R^2 = -0.026$ to $R^2 = 0.018$ — a substantial relative improvement, though SVR remained the weakest model overall, suggesting the RBF kernel is poorly suited to this sparse, high-dimensional one-hot-encoded feature space at the sample size used. Tuning produced

negligible change for Ridge and the Decision Tree, indicating the untuned defaults were already close to optimal for this dataset.

D. Feature-Subset Removal

Table 5 shows that removing `company_location` causes the largest drop in R^2 ($0.220 \rightarrow 0.155$), followed by removing `role_family` ($0.220 \rightarrow 0.176$). Removing `company_size` has almost no effect ($0.220 \rightarrow 0.219$). Using `experience_level` alone, the single feature most people intuitively associate with salary, explains only $R^2 = 0.105$, roughly half the variance explained by the full feature set, confirming that job role and geography are at least as important as seniority for this dataset.

Table 5. Feature-subset ablation (Ridge Regression)

Feature subset	MAE (\$)	R^2
All features	50,993	0.220
Without <code>role_family</code>	52,650	0.176
Without <code>company_location</code>	53,864	0.155
Without <code>company_size</code>	51,002	0.219
Experience level only	55,679	0.105

VII. Discussion

Linear Regression, Ridge Regression, and Decision Tree all perform the same ($R^2 \approx 0.22$). This shows that the simple linear pattern that these models use is the best that the available features can explain, especially because self-reported salary data is noisy. Ridge does slightly better than Linear Regression because it uses regularisation to lower variance caused by moderate multicollinearity that comes from one-hot encoding 97 company locations. The scaling comparison shows evidence for practitioners: scaling does not add cost or help Linear Regression or Decision Tree, but it is essential for neural network regressors on this mixed numeric/categorical data. If scaling is left out, the MLP falls to random performance ($R^2 = 0.007$). This drop could be mistaken for a modelling failure of a preprocessing mistake. SVR keeps performing even after tuning. This shows a problem with kernel methods on sparse high-cardinality one-hot feature spaces. The Euclidean distance used by the RBF kernel does not match the nature of the data. Using target or

frequency encoding for variables could improve SVR in the future. Removing feature subsets shows a result: job role and geographic location together explain about twice as much salary variance as experience level alone. This matters for salary-benchmarking tools, which often give much weight to years of experience and too little to role specialisation and location-based cost-of-living or market effects. From a practitioner’s point of view the results also give a rule: when a quick easy-to-understand salary-estimation model is needed from this type of tabular job-attribute data, a regularised linear model (Ridge) is a good default. It matches the accuracy of complex learners (Decision Tree, MLP) trains the fastest (Table 2) and is the easiest to interpret because of its coefficients. If more accuracy is needed, trying non-linear or ensemble methods (like Random Forest or Gradient Boosting) is a next step because earlier studies on similar data found ensemble tree methods to be competitive or better, than single decision trees.

Limitations

This study used a subsample of 10,000 rows from the full dataset, which had 71,913 rows, to make SVR training manageable. Tree-based models and linear models that handle datasets well could perform even better with the full dataset. The overall R^2 ceiling is around 0.22, which means a lot of the variation in salary is not explained by the variables we have. Factors like the employer, negotiated equity or bonus details, and individual candidate qualifications are not included in the dataset. The `company_location` variable groups the 8 countries plus an "Other" category, which reduces geographic detail but keeps the feature space manageable.

VIII. Conclusion

This study compared five machine learning regression algorithms for predicting AI/data-industry salaries from job- and employment-related information. After an adjustment of hyperparameters, Ridge Regression gave the best overall accuracy with a test R^2 of 0.220 and a mean absolute error of \$50,993. Linear Regression and a Decision Tree Regressor had performance. A focused comparison showed that scaling the input features matters a lot for neural-network regressors, improving the MLP’s R^2 from

0.007 to 0.215. Scaling does not change the results for linear and tree-based models. This directly answered the research question of the study. An ablation of feature subsets revealed that the role family and the company location predict AI/data-industry salaries better than experience level alone. Future research could try target encoding for high-cardinality variables, use ensemble methods such as Random Forest and Gradient Boosting, and run the linear and tree-based models on the full 71,913-row dataset, which does not need subsampling.

AI Usage Statement

ChatGPT was used to assist with: drafting the few sections of paper, such as generating the formatting for this manuscript, and locating and summarising related-work citations.

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